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MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)
Second Semester, March, 2025

Paper Code: BS 112**Subject: Applied Mathematics II****Time: 1½ Hrs.****Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q.No.	Question	Max. Marks	CO(s)
1. (a)	Find the modulus and argument of $i^{\ln(1+i)}$.	2.5	1
(b)	Verify whether the function $u(x, y) = 2xy + 3xy^2 - 2y^3$ is harmonic or not.	2.5	1
(c)	Determine the residue at the pole for $f(z) = (1 - e^{2z}) / z^4$.	2.5	2
(d)	Evaluate $\oint_C [dz / (z^2 + 9)]$, where C: $ z - 3i = 4$.	2.5	2
2. (a)	Find the analytic function $f(z) = u + iv$ where $u - v = (x - y)(x^2 + 4xy + y^2)$ and $z = x + iy$.	6	1
(b)	Evaluate $\oint_C (z - z^2) dz$, where C is the upper half of the circle $ z - 2 = 3$.	4	1
3. (a)	Find the Laurent Series of $f(z) = (7z^2 + 9z - 18) / (z^3 - 9z)$ about $z = 0$.	6	2
(b)	Find the image of $ z - 3i = 3$ under the transformation $w = 1 / z$ in w plane.	4	2
4. (a)	Evaluate $\int_0^{2\pi} [4 / (5 + 4 \sin \theta)] d\theta$.	5	2
(b)	Show that the function $f(z) = x^2 + iy^3$ is nowhere analytic.	5	1

**** End of Question Paper ****